

Total No. of Questions : 10]

SEAT No. :

PD5322

[Total No. of Pages : 2

[6404] 1523

B.E. (Computer Engineering)
EMBEDDED & REAL TIME OPERATING SYSTEMS
(2015 Pattern) (Semester - VIII) (410252 C)(Elective -III)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answers Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Draw neat diagram whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain in detail different characteristics of embedded systems. [5]
b) Draw the block diagram of an embedded system and describe the hardware units of an embedded system. [5]

OR

- Q2)** a) Explain types of serial communication with examples. [5]
b) Write a short note on ARM bus. [5]

- Q3)** a) What are the main features of CAN2.0 Bus standards? [5]
b) Describe the parallel protocols PCI, PCI-X. [5]

OR

- Q4)** a) Describe and compare RS232C and SDIO Devices. [5]
b) Name and Explain Power Management modes of Bluetooth. [5]

- Q5)** a) How the Precedence constraint and data dependency decides in real time task? [7]
b) Explain Earliest Deadline first and Rate Monotonic Algorithm in detail. [9]

OR

P.T.O.

- Q6) a)** Use EDF Scheduler for scheduling following periodic task T1 and T2. Perform schedulability check and comment on whether given task is schedulable and schedule produced is feasible. [9]

Task	Burst Time	Deadline	Period
T1	3	7	20
T2	2	4	5
T3	2	8	10

- b) Differentiate hard verses soft real time system and their time constraints. [7]
- Q7) a)** What is Priority Inversion? What is deadlock? With appropriate example prove that priority ceiling algorithm avoids deadlock. [9]
- b) What is Semaphore? How does it help in resource sharing in RTOS kernel? [7]

OR

- Q8) a)** What is shared data problem? Explain any two methods to avoid it. [9]
- b) Explain with example Resource conflicts and blocking. [7]
- Q9) a)** Explain in detail software development process in Embedded System. [9]
- b) Describe in details the features of QNX Neutrino. [9]

OR

- Q10)a)** Explain with example Validation and debugging in an embedded system. [9]
- b) Write in detail resource access control and synchronization in Real-time Operating system. [9]

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